M6A1: Problem Set — Kossiakoff Chapter 13, Problem 13.2

EN.655.662 Introduction to Healthcare Systems Engineering | [Your Name] | [Date]

Problem 13.2 — What specific activities in the advanced development phase sometimes cause it to be referred to as a "risk reduction" phase? Give an example of each activity considering a real or hypothetical system.

System used for the examples: the remote cardiac care system — an implanted or worn cardiac device, a patient-home desk transceiver, a cellular transport link, a clinical computer stack, and a clinician-facing clinical station.

Advanced development sits between concept definition and engineering design. Concept definition ends with a chosen concept; engineering design begins an expensive and largely irreversible commitment to a particular physical realization. Advanced development is the last interval in which the organization can still change its mind cheaply, and its product is therefore not a design but **evidence**. Seven activities produce that evidence, and each retires a distinct class of uncertainty. Together they are why the phase is called risk reduction.

1. Requirements Analysis and Flow-Down

Operational requirements are decomposed into system and subsystem requirements that are measurable, verifiable and traced to the functions they serve. This retires **requirements risk** — the failure mode in which a requirement proves ambiguous or untestable only after a design has been built to satisfy it. *Example:* the operational requirement that the transponder have "a minimum communications range of 40 ft" is not directly testable as written. Flowing it down against the Set Up Cardiac Comms function yields requirements that are: power-up in no more than 5 s, an activation force of no more than 1 N, and power confirmation by a 55 dB, 2 Hz, 200 ms audio chirp. The ambiguity is removed before anyone builds to it.

2. Technology Identification and Maturation

Each element is assessed for maturity, conventionally by technology readiness level, and the immature elements are matured deliberately. This retires **technology risk** and, just as importantly, tells the programme where to spend. *Example:* the remote cardiac system is not uniformly mature. The clinical computer stack and the cellular transport are effectively commercial off-the-shelf at TRL-9; the desk transceiver and the clinical station sit near TRL-7; end-to-end operation with a live cardiac device and a patient is the least mature at roughly TRL-6. The assessment shows that risk is concentrated in the integration, not in any individual box, and redirects the prototyping budget accordingly.

3. Prototyping

Prototypes are built to answer questions rather than to be the product — explorative prototypes surface requirements with users, experimental prototypes test whether the planned system will be adequate, and evolutionary prototypes are refined into the delivered system. This retires **feasibility and usability risk**. *Example:* a field prototype of the desk transceiver and its cellular link — a radio, an antenna and a data logger, with no enclosure — placed in a sample of real patient homes. It answers a question no amount of analysis can: whether coverage holds in a basement bedroom or a steel-framed building. Keeping it low fidelity matters, because a polished prototype consumes schedule and can quietly anchor the eventual design to its particular form.

4. Modeling, Simulation and Analysis

Where a prototype is premature or expensive, an explicit model bounds behaviour analytically and supports early, iterative verification rather than deferring all of it downstream. This retires **performance risk**. *Example:* a link-budget and queueing model of the cellular transport, evaluated against the

requirement that the data rate be twice that of the cardiac device. The model establishes that average throughput and latency satisfy the requirement comfortably — and, read honestly, also establishes what it cannot address, namely intermittent dropout in a specific home. That boundary is what justifies activity 3, and the pairing shows why modeling and prototyping are complements rather than substitutes.

5. Trade Studies

Alternatives are identified and compared against measures of feasibility, performance, cost, schedule and risk, with the estimates drawn from modeling, prototyping or surrogates. This retires **selection risk**, the commitment to an architecture that turns out to be a dead end. *Example:* the transport architecture is not settled by assumption. Cellular, a home broadband gateway, and a Bluetooth link relayed through the patient's smartphone are scored against coverage in the target population, recurring cost per patient, patient burden at setup, and dependence on equipment the patient owns and may replace. The smartphone option is cheapest and fails worst on an elderly cohort — a conclusion worth reaching on paper rather than after a fielded pilot.

6. Interface Definition and Developmental Test Planning

Interfaces between elements are specified, and the test and evaluation master plan, test equipment and test software are developed. This retires **integration risk**, which dominates in a distributed system, and **verification risk**, the late discovery that a requirement cannot be tested as written. *Example:* the integration is sequenced so that each step introduces exactly one new unknown — confirm the desk transceiver alone; isolate the clinical computer stack; isolate the clinical station; combine station and stack; add the cellular link for a partial end-to-end test; run end-to-end without the cardiac device; then with it; then with a patient. When a test fails, the cause is unambiguous, because only one thing changed.

7. Baselining the Validated System Specification

The phase exits by baselining a system specification that has been demonstrated to be achievable, so that engineering design proceeds against a stable target. This retires **baseline risk**. *Example:* the dropout behaviour discovered in activity 3 is resolved before baselining, by adding a store-and-forward buffer to the desk transceiver with a defined retention window and reconnection behaviour, and flowing the corresponding requirement down into the specification.

Why These Activities Make It a "Risk Reduction" Phase

Two arguments converge. The first is economic: by the close of advanced development the great majority of total product cost is already committed even though very little has been spent, so a change made here is cheap and the identical change made later is not. Had the cellular dropout of activity 3 instead surfaced after engineering design, the response would have been a hardware respin of a device already in production, a regulatory change assessment, and a clinical safety case built around the possibility that an arrhythmia event was never transmitted. The engineering work is much the same; the cost and the patient consequence differ by orders of magnitude. The second argument is epistemic: every other phase produces an artifact, while this one produces knowledge. A programme that skips advanced development has not avoided its uncertainty — it has only moved the discovery of that uncertainty into a phase where acting on it is expensive.